Prime-Indexed Primes: Exact Counting and Asymptotics

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Abstract

For the prime-indexed primes $s_n = p_{p_n}$ (OEIS A006450), the exact counting function is $\pi(\pi(x))$. As consequences of the prime number theorem, $s_n \sim n(\log n)^2$ and the number of prime-indexed primes not exceeding x is asymptotic to $x/(\log x)^2$. Their reciprocal series converges. These familiar properties are proved here with the precise dependency on the prime number theorem stated.

Let p_n denote the n-th prime ($p_1 = 2$) and $\pi(x) = \#\{p \le x : p \text{ prime}\}$. Define

$$s_n = p_{p_n} \quad (n \ge 1), \qquad S(x) = \#\{n \ge 1 : s_n \le x\}.$$

OEIS A006450 begins $3, 5, 11, 17, 31, 41, 59, 67, 83, 109, \ldots$.

Theorem 1 (Exact count)**.** *For every real $x \ge 0$,*

$$S(x) = \pi(\pi(x)). \tag{1}$$

Proof. Because the primes are strictly increasing, the condition $p_{p_n} \le x$ is equivalent to $p_n \le \pi(x)$. The number of primes p_n not exceeding $\pi(x)$ is exactly $\pi(\pi(x))$. □

Theorem 2 (Density and growth from the PNT)**.** *As $x \to \infty$ and $n \to \infty$ respectively,*

$$S(x) \sim \frac{x}{(\log x)^2}, \qquad s_n \sim n(\log n)^2.$$

Consequently, the proportion of primes $\le x$ which have prime index is asymptotic to $1/\log x$ and tends to 0.

Proof. We use the standard prime number theorem (PNT) in either equivalent form

$$\pi(x) \sim \frac{x}{\log x}, \qquad p_n \sim n \log n. \tag{2}$$

In particular $\pi(x) \to \infty$ and

$$\log \pi(x) = \log x - \log\log x + o(1) \sim \log x.$$

Using Theorem 1 and the PNT twice gives

$$S(x) = \pi(\pi(x)) \sim \frac{\pi(x)}{\log \pi(x)} \sim \frac{x}{(\log x)^2}.$$

Likewise $p_n \sim n \log n$, whence $\log p_n \sim \log n$, and

$$s_n = p_{p_n} \sim p_n \log p_n \sim n(\log n)^2.$$

Finally $S(x)/\pi(x) \sim 1/\log x$ by (2). □

Corollary 1 (Convergence of the reciprocal series)**.**

$$\sum_{n=1}^{\infty} \frac{1}{s_n} < \infty.$$

Proof. Theorem 2 supplies n_0 and $c > 0$ such that $s_n \geq cn(\log n)^2$ for every $n \geq n_0$. The comparison series $\sum_{n\geq 2} 1/(n(\log n)^2)$ converges by Cauchy condensation: for $a_n = 1/(n(\log n)^2)$, the condensed series is

$$\sum_{k\geq 1} 2^k a_{2^k} = \frac{1}{(\log 2)^2} \sum_{k\geq 1} \frac{1}{k^2} < \infty.$$

Adding finitely many initial terms does not affect convergence. □

Finite checks. From the first ten prime-indexed primes one obtains $S(100) = 9$; the exact identity also gives $\pi(\pi(100)) = \pi(25) = 9$. In addition, the conjectured strict decrease of $s_n^{1/n}$, mentioned on the OEIS page, was numerically checked here for $1 \leq n \leq 1000$, but this finite check is not a proof of that conjecture.

Scope, provenance and verification

The exact counting identity, asymptotic term growth, and reciprocal-series convergence are already recorded on OEIS A006450. The asymptotic theorems rely on the *established* prime number theorem, not an elementary proof of it. The separate conjectured strict decrease of $s_n^{1/n}$ remains beyond the scope of the note; no novelty is asserted. *AI authorship notice.* This document was prepared by an AI system. The displayed proofs were algebraically reviewed and finite examples checked computationally, but there has been no independent human peer review. Numerical checks do not substitute for a proof. No research-priority or originality claim is made.

References

1. OEIS Foundation, A006450: Prime-indexed primes (counting formula, asymptotics, reciprocal sum, and a conjecture).
2. NIST Digital Library of Mathematical Functions, §27.12, Asymptotic Formulas: Primes (PNT and $p_n \sim n \log n$).
3. C. Axler, "New Estimates for the nth Prime Number," *Journal of Integer Sequences* **22** (2019), Article 19.4.2.